

Design of an Ultra-Low Noise, High Gain Ku-Band CMOS LNA

A SNIM-Based Approach on 65nm Bulk CMOS

Samarth Walse (251040092) Purwansh Sahu (251040079)

Indian Institute of Technology Kanpur

April 20, 2026

Outline

- 1 Project Objectives & Benchmarking
- 2 SNIM Theory & Mathematical Formulation
- 3 Design Strategy & Component Selection
- 4 Design Analysis: Power vs. Noise Trade-off
- 5 Simulation Results & Comparison
- 6 Conclusion
- 7 References

Project Objectives & Benchmarking

The Goal

Design a fully integrated Ku-Band LNA (10.7 – 12.75 GHz) utilizing a TSMC 65-nm bulk CMOS process to achieve simultaneous 50 Ω input/output matching and an ultra-low noise figure of ≤ 1.1 dB.

Commercial Benchmark: Renesas F6931 (SOI)

Achieving SOI-level noise performance on a lossy bulk CMOS substrate requires aggressive architectural innovations to meet these targets:

- **Center Freq:** 11.725 GHz
- **Gain (S_{21}):** ≥ 24 dB
- **Supply (V_{DD}):** 1.2 V
- **Noise Figure:** ≤ 1.1 dB
- **Power:** ≤ 11.4 mW/ch

The Core Challenge: The Lossy Gate Inductor

The Standard LNA Paradigm

Traditional LNAs employ inductive source degeneration (L_s) and a series gate inductor (L_G) to achieve a real $50\ \Omega$ input match (Z_{in}):

$$Z_{in}(j\omega) = R_G + \frac{g_m}{C_{gs}} L_s + j\omega(L_G + L_s) - \frac{j}{\omega C_{gs}} \quad (1)$$

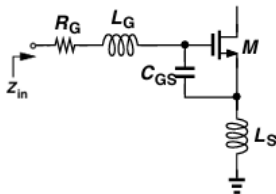


Figure 1: CS Stage with Inductive Degeneration

The Core Challenge: Our Directive

The Noise Penalty in Bulk CMOS

In 65nm bulk CMOS, on-chip spiral inductors exhibit exceptionally low quality (Q) factors due to substrate losses. The parasitic resistance (R_G) of a series L_G contributes severe thermal noise directly at the most sensitive node in the receiver chain.

Our Directive

Eliminate the series L_G entirely. To meet the stringent 1.1 dB Noise Figure benchmark of the commercial SOI component, we must achieve Simultaneous Noise and Input Matching (SNIM) purely through device physics and parallel components, completely avoiding lossy series matching networks.

SNIM: The Conflict and Ultimate Condition

Two Competing Matching Conditions

LNA design requires balancing two traditionally conflicting input goals:

- **Noise Match:** $Z_s = Z_{opt}$ (achieves minimum noise figure, F_{min}).
- **Power Match:** $Z_s = Z_{in}^*$ (maximizes power transfer, zero reflection).

The Ultimate SNIM Requirement

For a standard $50\ \Omega$ system ($Z_s = 50\ \Omega$, purely real), Simultaneous Noise and Impedance Matching (SNIM) strictly requires both parameters to align:

$$Z_{in} = Z_{opt}^* = Z_s^* = 50\ \Omega \implies \text{Re}\{Z_{in}\} = \text{Re}\{Z_{opt}^*\} = 50\ \Omega \quad (2)$$

The Challenge: Making both the complex impedances and their real parts perfectly equal to $50\ \Omega$ simultaneously.

Mathematical Formulation of SNIM: Device Physics

The MOSFET Characteristics (with L_s)

Based on the rigorous hybrid- π noise model, Z_{opt}^* is governed by intrinsic parasitics, while Z_{in} is heavily modified by the source degeneration inductor (L_s):

$$Z_{opt}^* \approx \frac{1}{\omega(C_{gs} + C_{gd})} \sqrt{\frac{\alpha}{\gamma g_m} g_m^2 R_g} - j \frac{1}{\omega(C_{gs} + C_{gd})} \quad (3)$$

$$Z_{in} \approx \left(R_g + \frac{g_m}{C_{gs}} L_s \right) + j \left(\omega L_s - \frac{1}{\omega C_{eff}} \right) \quad (4)$$

(Notice that the lossless source inductor L_s introduces a real resistive term to Z_{in} without fundamentally altering the noise parameter Z_{opt}^ .)*

Mathematical Formulation of SNIM: Device Physics

The Fundamental Discrepancy

For a conventionally sized transistor, the optimum noise impedance and input impedance are vastly mismatched ($\text{Re}\{Z_{opt}^*\} \gg \text{Re}\{Z_{in}\}$), and neither naturally sits at $50\ \Omega$.

Fixing Z_{opt}^* via Device Scaling

Because $\text{Re}\{Z_{opt}^*\}$ is inversely proportional to the total capacitance $\omega(C_{gs} + C_{gd})$, we can manipulate the optimum noise impedance solely by increasing transistor sizing to get near $50\ \Omega$ impedance.

Fixing Z_{in} Via L_s

To force a $50\ \Omega$ power match, we can typically use L_s to control the real part of Z_{in} and bring it near $50\ \Omega$.

SNIM: Sizing for Optimum NF_{min}

The Sizing Trade-off

At 12 GHz, increasing device size initially reduces NF_{min} before excessive parasitic capacitance (C_{gs} , C_{gd}) and gate resistance (R_g) begin to degrade it. Simulation yields a distinct optimal width.

The Sizing Selection

We selected $W = 240 \mu\text{m}$ because it sits precisely near the absolute minimum of the NF_{min} curve ($\sim 0.54 \text{ dB}$). However, this massive size also serves a crucial second purpose for our impedance matching.

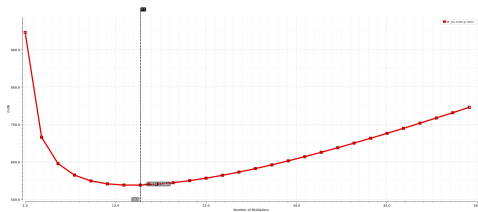


Figure 2: Simulated minimum noise figure (NF_{min}) at 12 GHz for the cascode structure, at $V_{dd} = 1.2 \text{ V}$, $V_g = 0.5 \text{ V}$.

SNIM: Aligned $\text{Re}\{Z_{opt}^*\}$ and Aligning $\text{Re}\{Z_{in}\}$

Our Aggressive Sizing Strategy

The large $240\text{ }\mu\text{m}$ choice also helps us increase the total capacitance ($C_{gs} + C_{gd}$) required to align the real part of Z_{opt}^* . By deliberately increasing this capacitance, we drive the real part of Z_{opt}^* down to $40\text{ }\Omega$ at 12 GHz :

$$\text{Re}\{Z_{opt}^*\}|_{W=240\mu\text{m}} \approx 40\text{ }\Omega \quad (5)$$

Managing the Input Impedance (Z_{in})

While this massive size aligns Z_{opt}^* , it drastically alters the physical Z_{in} :

- **Reducing R_g :** The highly parallelized layout (12 fingers and 20 multipliers) reduces the physical gate resistance to a negligible $\sim 1\text{ }\Omega$.
- **Base Resistance:** Instead of using a noisy series L_G , the Stage 1 source inductor ($L_s \approx 100\text{ pH}$) provides a clean, noiseless base resistance ($\text{Re}\{Z_{in}\} \approx \frac{g_m}{C_{gs}} L_s \approx 50\text{ }\Omega$).

SNIM: Susceptance Cancellation via L_{ESD}

Step 1: The Constant Admittance Circle

As shown on the Smith chart locus, the massive size of the input transistor ($W = 240 \mu\text{m}$) inherently moves both the optimum noise impedance (Z_{opt}) and the conjugated gate impedance (Z_{in}^*) directly onto the constant admittance circle.

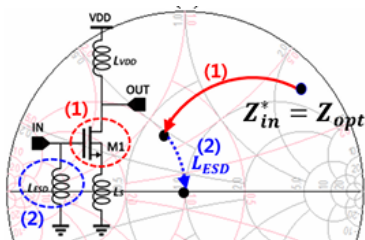


Figure 3: Locus of Z_{opt} and Z_{in}^* shifting to the constant admittance circle due to the massive device sizing. *ref[3]*

SNIM: Susceptance Cancellation via L_{ESD}

Step 2: Admittance Match & Calculation

The massive device approximately aligns the real admittance ($\text{Re}\{Y_{in}\} \approx 1/50 \text{ S}$) but introduces a large parasitic capacitance. A shunt inductor (L_{ESD}) is added to provide the exact negative susceptance required to cancel it.

Design Calculation (at $f_0 = 11.725 \text{ GHz} \Rightarrow \omega \approx 73.67 \text{ Grad/s}$):

- Extracted device capacitance: $C_p \approx 315 \text{ fF} \Rightarrow B_{device} \approx 23.2 \text{ mS}$
- Cancellation requirement: $j\omega C_p - j(\omega L_{ESD})^{-1} \approx 0$
- Theoretical inductor: $L_{ESD} \approx \frac{1}{\omega^2 C_p} \approx 585 \text{ pH}$

(Final adjustments after adding biasing networks and the second stage shifted our final tuned value to $L_s \approx 657 \text{ pH}$.)

SNIM: Final Conclusion & Dual Functionality

Achievement of SNIM

Conclusion: We achieved Simultaneous Noise and Impedance Matching ($Z_{in} \approx Z_{opt}^* \approx 50 \Omega$) purely through device sizing and shunt tuning, entirely eliminating the thermal noise penalty of a lossy series L_G .

ESD Protection

Dual Functionality: Furthermore, because L_{ESD} is a grounded shunt element, it provides a robust, low-resistance DC discharge path to protect the thin gate oxide of the massive input transistor from Electrostatic Discharge (ESD) events.

Stage 1: The SNIM Core Sizing

Stage 1 Transistors (M1 & M2) – 240 μm

- **Sizing:** Unit finger width $W_f = 1 \mu\text{m}$, with $nf = 12$ fingers and $m = 20$ multipliers (240 μm total).
- **Parasitic Mitigation:** This highly parallelized (12×20) layout lowered the physical gate resistance (R_g), which would otherwise degrade F_{min} .

Input Matching Network (L_{ESD} & C_{DC})

- L_{ESD} (**$\sim 657 \text{ pH}$**): Provides the exact negative susceptance to cancel C_{gs} . Crucially, it also acts as a safe, low-impedance DC discharge path to ground, providing robust **ESD Protection** for the 65nm oxide without adding series thermal noise.
- C_{DC} (**4.16 pF**): Blocks DC bias while fine-tuning the real part of the impedance match.

Stage 2: Linearity & Stability Core Sizing

The Cascode Devices (M3 & M4) – $192\ \mu\text{m}$

- **Sizing:** Scaled down to unit width $W_f = 1\ \mu\text{m}$, with $nf = 12$ fingers and $m = 16$ multipliers ($192\ \mu\text{m}$ total).
- **Rationale:** Stage 2 is sized smaller to conserve DC power, as Friis's formula dictates that Stage 1 dominates the Noise Figure.

Inductive Source Degeneration (L_S)

An inductor ($L_S \approx 72\ \text{pH}$) is placed at the source of M3:

- **Linearization:** Provides series-series negative AC feedback, linearizing G_m and boosting Output IP3 to 11.46 dBm.
- **Stability:** Generates a real resistive term ($R_{in} \approx \frac{g_m}{C_{gs}} L_S$), ensuring the Rollett Stability Factor (K_f) is > 1 .

Stage 1 to Stage 2: Interstage Matching & Peaking

Stage 1 Drain Peaking Inductor (L_{d1})

To maximize the voltage gain of the first stage, we employ a shunt peaking inductor ($L_{d1} \approx 547$ pH) at the drain of M2.

- **Resonance:** L_{d1} is sized to resonate out the combined parasitic capacitances at this node (C_{dd} of M2, and C_{gs} of M3) exactly at our 11.725 GHz center frequency.

Interstage AC Coupling (C_{inter})

A 400 fF series capacitor is placed between the stages:

- **Maximizing Voltage Swing:** C_{inter} forms a capacitive voltage divider with the input capacitance of M3 (C_{gs3}). By sizing it to 400 fF ($C_{inter} > C_{gs3}$), we ensure maximum AC voltage transfer to the Stage 2 gate.

Stage 2 Output: Bandwidth & S_{22} Matching

Stage 2 Gain Flattening (L_{d2} & R_{d2})

The Stage 2 drain utilizes a resonant inductor ($L_{d2} \approx 650$ pH) combined with a $300\ \Omega$ parallel resistor.

- **Flattening the gain:** The circuit naturally wants to create a sharp, narrow spike in power. The resistor acts like a damper, trading a little bit of that maximum peak height to keep the signal flat and steady across the whole range we need.

Output Matching Network (C_{series} & L_{series})

- **DC Blocking ($C_{series} = 115$ fF):** Isolates the $1.2\text{V } V_{DD}$ drain bias to prevent DC shorts to the external load.
- **Impedance Tuning ($L_{series} \approx 694$ pH):** Adds precise positive reactance to tune the net series arm to exactly $X_{net} \approx -67\ \Omega$. This completes the $\sim 140\ \Omega$ to $50\ \Omega$ step-down L-match.

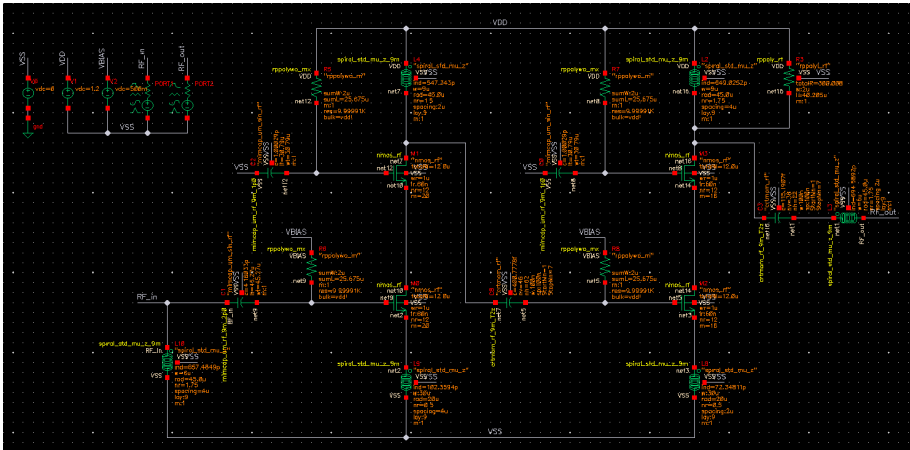


Figure 4: Complete schematic of the proposed Ku-Band LNA, featuring the SNIM core, cascode second stage, and reactive matching networks.

Design Analysis: The Power vs. Noise Trade-off

The Bulk CMOS vs. SOI Reality

The Renesas F6931 achieves 11.4 mW using Silicon-on-Insulator (SOI) technology, which inherently possesses lower parasitic capacitances and superior substrate isolation compared to our 65nm Bulk CMOS process.

Prioritizing the 1.1 dB Noise Figure

To break the 1.1 dB NF barrier on a bulk substrate, we deliberately implemented the high-current, large-width SNIM topology operating at 1.2V.

- **The Penalty:** DC Power consumption increased to 51.44 mW.
- **The Benefit:** Achieved an exceptional NF of 1.06 dB, a massive peak gain of 30.32 dB, and an Output P_{1dB} of 2.43 dBm (exceeding the -1 dBm target).

Performance Comparison: Noise Figure (NF)

Our Design (PVT Corners)

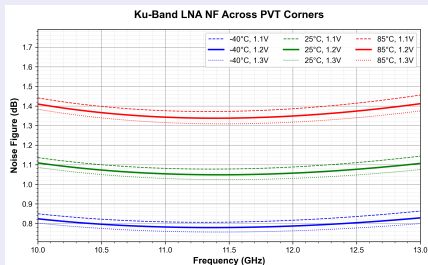


Fig: Simulated NF (9 Corners)

Commercial Target

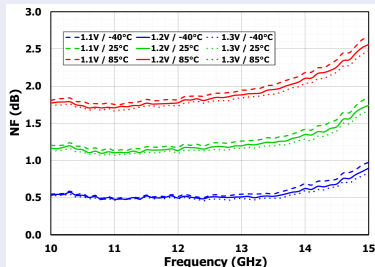


Fig: Reference Commercial NF

- **Observation:** The large-device SNIM topology achieves a minimum NF of **1.06 dB** (11.725 GHz). Even at the worst-case 85°C corner, performance rivals commercial benchmarks.

Performance Comparison: Forward Gain (S_{21})

Our Design (PVT Corners)

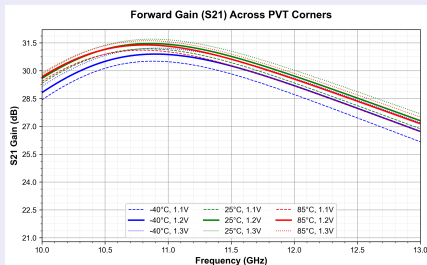


Fig: Simulated S_{21} (9 Corners)

Commercial Target

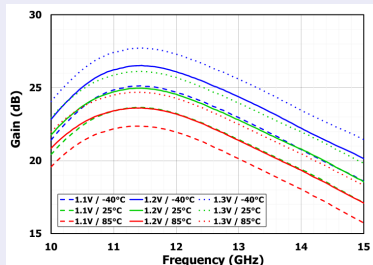


Fig: Reference Commercial S_{21}

- Observation:** The 2-stage cascode with shunt peaking delivers a massive peak gain of **30.32 dB**. The 300 Ω parallel resistor ensures wideband stability across PVT variations.

Performance Comparison: Input Return Loss (S_{11})

Our Design (PVT Corners)

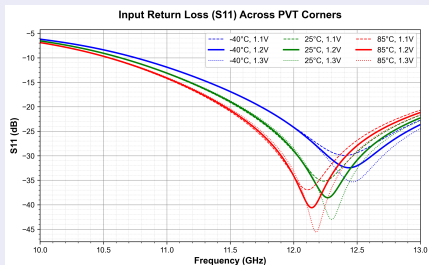


Fig: Simulated S_{11} (9 Corners)

Commercial Target

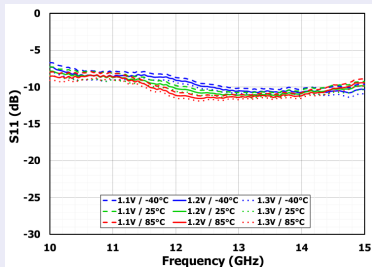


Fig: Reference Commercial S_{11}

- **Observation:** The dual-function shunt L_{ESD} perfectly cancels input capacitance, yielding a deep match of **-22.62 dB** at center frequency and remaining < -10 dB across the Ku-band.

Performance Comparison: Output Return Loss (S_{22})

Our Design (PVT Corners)

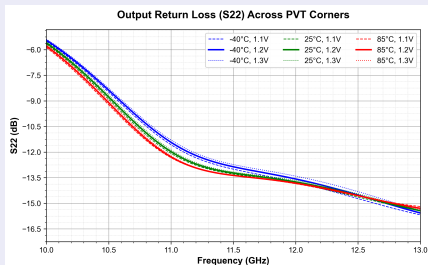


Fig: Simulated S_{22} (9 Corners)

Commercial Target

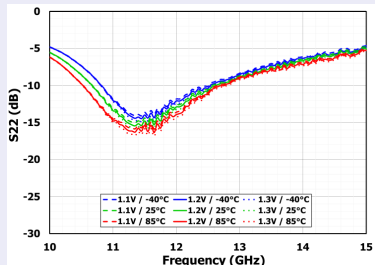


Fig: Reference Commercial S_{22}

- Observation:** The series L-C network effectively steps down the $\sim 140 \Omega$ drain impedance to 50Ω . The design reliably holds $S_{22} \leq -12$ dB under severe process and temperature stress.

Performance Comparison: Reverse Isolation (S_{12})

Our Design (PVT Corners)

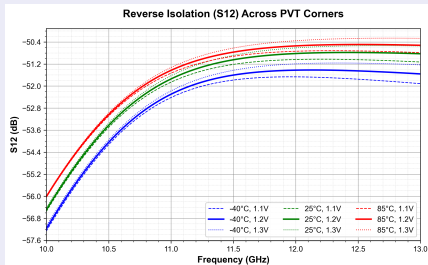


Fig: Simulated S_{12} (9 Corners)

Commercial Target

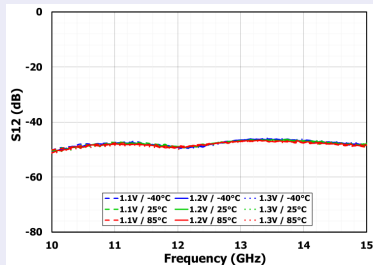


Fig: Reference Commercial S_{12}

- **Observation:** The cascode topology drastically cuts Miller feedback capacitance (C_{gd}). This guarantees excellent reverse isolation (< -45 dB), ensuring unconditional stability.

Performance Verification: Stability

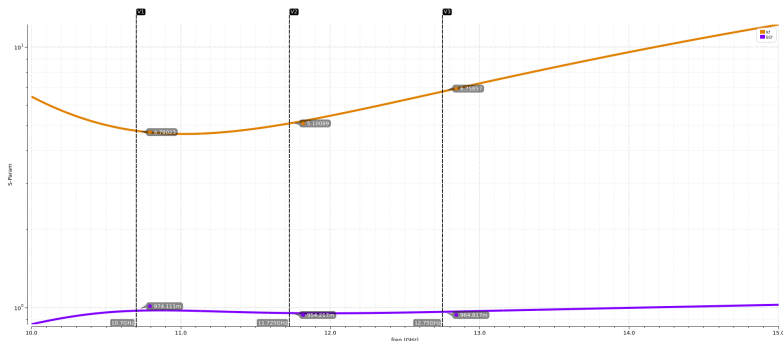


Fig: Simulated Stability Factors (K_f & B_{1f})

- **Observation:** The LNA is unconditionally stable across the entire Ku-band ($K_f > 1$, $B_{1f} < 0$).

Linearity Performance: Output IP3

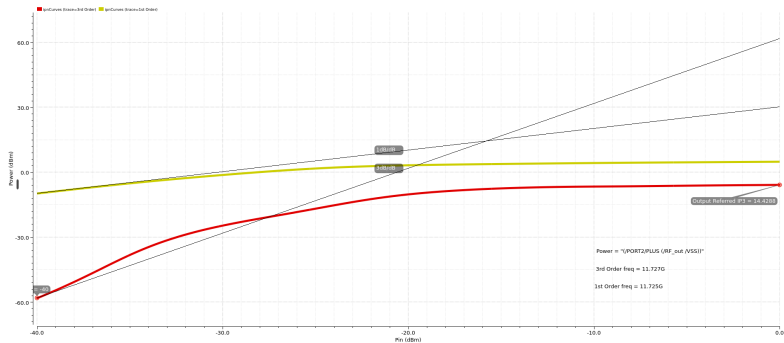


Fig: Output Third-Order Intercept Point (OIP3)

- **Observation:** Operating the large-width SNIM topology with higher current density yields a simulated **OIP3 of 14.43 dBm**, successfully exceeding the 8 dBm commercial benchmark of the F6931.

Power Handling: Output 1-dB Compression (OP_{1dB})

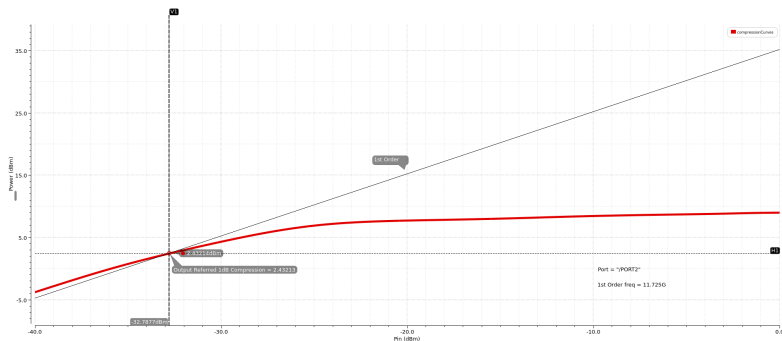


Fig: Output 1-dB Compression Point (OP_{1dB})

- **Observation:** The design achieves an OP_{1dB} of **2.43 dBm**, successfully exceeding the commercial datasheet target of **-1.0 dBm**.

Summary of Specifications & Power Trade-off

Parameter	Renesas F6931	Simulated (1.2V)	Simulated (1.0V)
Frequency Band	10.7 – 12.75 GHz	10.7 – 12.75 GHz	10.7 – 12.75 GHz
Supply Voltage (V_{DD})	1.2 V	1.2 V	1.0 V
Gain (S_{21})	24 dB	30.32 dB	29.54 dB
Noise Figure (NF)	1.1 dB	1.06 dB	1.13 dB
Output P_{1dB}	-1 dBm	2.43 dBm	1.65 dBm
Output IP3 (OIP3)	8 dBm	14.43 dBm	13.56 dBm
Input Match (S_{11})	≤ -10 dB	-22.62 dB	-22.98 dB
Output Match (S_{22})	≤ -12 dB	-13.45 dB	-13.61 dB
Reverse Isolation (S_{12})	-48 dB	-50.9 dB	-51.28 dB
Power Consumption	11.4 mW	51.44 mW	36.12 mW

Table 1: Performance Benchmarking & Power Scaling (Values at 11.725 GHz)

Conclusion & Future Work

Conclusion

- Successfully designed a Ku-Band LNA exceeding commercial gain, linearity, and noise specifications on a 65nm Bulk CMOS process.
- Eliminated the lossy series gate inductor (L_G), mathematically relying on intrinsic device scaling (C_{gs}) for primary conductance matching.
- Achieved unconditional stability ($K_f = 4.78$) and excellent linearity through 1.2V operation and inductive source degeneration.

Future Work

- **Power Optimization:** Scale down the Stage 2 cascode to trade the excess 6.3 dB of gain margin for a significant reduction in the 51.44 mW power.
- **Post-Layout Verification:** Perform rigorous layout and quasi-3D-EM co-simulation (e.g., EMX) to extract routing parasitics and iteratively re-center the admittance matching network.

References



Renesas Electronics.

F6931: Dual-Channel Low Noise Amplifier 10.7 GHz to 12.75 GHz Datasheet.
Rev. 1.02, Nov. 2024.



C.-T. Fu, C.-N. Kuo, and S. S. Taylor.

Low-Noise Amplifier Design With Dual Reactive Feedback for Broadband Simultaneous Noise and Impedance Matching.

IEEE Trans. Microw. Theory Techn., vol. 58, no. 4, pp. 795–806, Apr. 2010.



H.-W. Choi, S. Choi, and C.-Y. Kim.

Ultralow-Noise Figure and High Gain Ku-Band Bulk CMOS Low-Noise Amplifier With Large-Size Transistor.

IEEE Microw. Wireless Compon. Lett., vol. 31, no. 1, pp. 60–63, Jan. 2021.



J.-T. Son, H.-W. Choi, and C.-Y. Kim.

Sub-6 GHz LNA Using Two-Stage SNIM With Series Interstage Inductor Based on 0.5- μm GaAs E-pHEMT Technology.

IEEE Microw. Wireless Technol. Lett., vol. 33, no. 9, pp. 1301–1304, Sept. 2023.